

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	07 July 2026
Team ID	
Project Name	Visualization tool for vehicle charge and range analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Visualization Tool for Electric Vehicle Charge and Range Analysis

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Guidelines:

- Import EV dataset (Excel/CSV) into Tableau.
- * Clean and preprocess the dataset before visualization.
- * Create interactive dashboards using charts, graphs, and KPIs.
- * Enable filtering by vehicle model, manufacturer, battery capacity, charging time, and year.
- * Store project files locally and maintain version control using GitHub.
- * Generate visual insights to support EV charge and range analysis.

Technical Architecture Diagram

Visualization Tool for Electric Vehicle Charge and Range Analysis

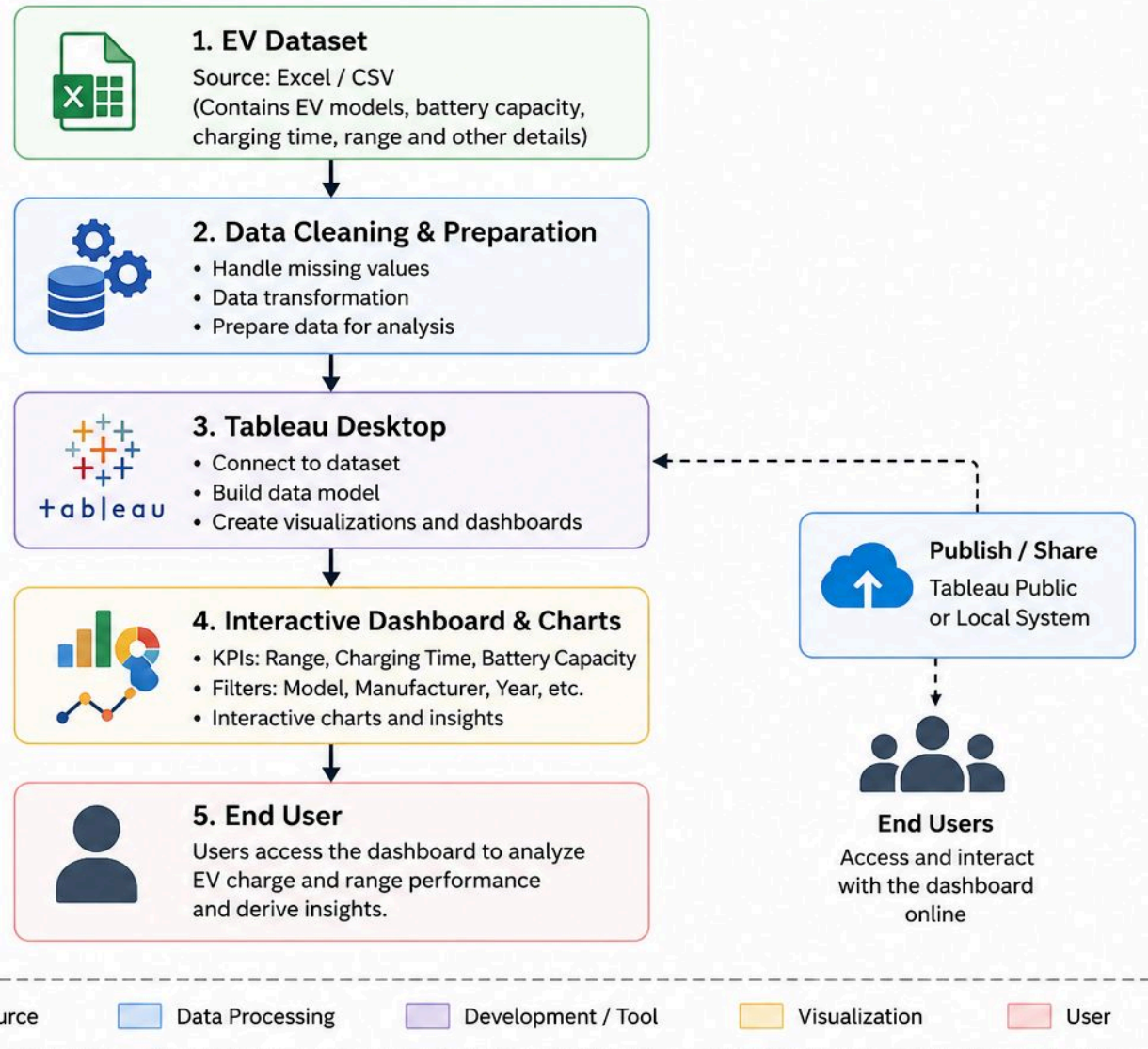


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for visualization	Tableau Dashboard
2.	Application Logic-1	Import EV dataset	Tableau desktop
3.	Application Logic-2	Data cleaning and transformation	Tableau desktop / Tableau prep
4.	Application Logic-3	Data visualization and dashboard creation	Tableau desktop
5.	Database	EV dataset	Excel(.xlsx) / CSV
6.	Cloud Database	Not Applicable	N/A
7.	File Storage	Store project files and dataset	Local storage / GitHub
8.	External API-1	Not Applicable	N/A
9.	External API-2	Not Applicable	N/A
10.	Machine Learning Model	Not Applicable	N/A

11.	Infrastructure (Server / Cloud)	Tableau dashboard deployment	Tableau Public / Local system
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Libraries and tools used	Tableau Public
2.	Security Implementations	Secure handling of project files	GitHub Repository / Local File Security
3.	Scalable Architecture	Dashboard supports larger EV datasets	Tableau
4.	Availability	Dashboard accessible online or locally	Tableau Public
5.	Performance	Fast loading of dashboard and filters	Tableau Optimized Dashboard

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>